

Door-Opening Alarm Using Hall Sensor



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This simple door-opening alarm uses a hall sensor to produce sound when someone opens the door. The circuit can be installed at a workshop, office or home for security purposes. The 3-terminal hall sensor is easily available in the market.

Circuit and working

The circuit diagram of the door-opening alarm is shown in Fig. 1. It is built around hall sensor MH183

(IC1), pnp transistor BC557 (T1), npn transistor BC547 (T2) and a few other components.

Hall sensor MH183 is a magnet-operated solid-state sensor. When the magnet is brought near the sensor, its output goes low. This low output is amplified by pnp transistor T1. npn transistor T2 inverts the operation. When T1 conducts, T2 does not conduct, and vice-versa.

The magnet should be fitted on moving part of the door and the sensor at the door frame. When the door is closed, the magnet comes close to the sensor and the alarm is off. But when door is opened, the alarm makes a sound.

The buzzer can be replaced with a 5V relay to control a mains alarm.

When the

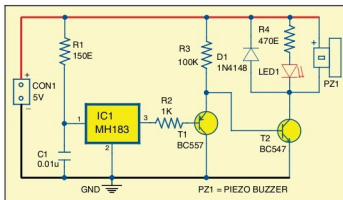


Fig. 1: Circuit diagram of the door-opening alarm

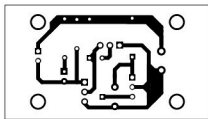


Fig. 2: Actual-size PCB of the door-opening alarm

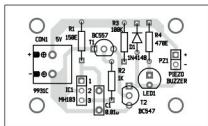


Fig. 3: Component layout of the PCB

PARTS LIST

Semiconductors:

- IC1 - MH183 hall sensor
- T1 - BC557 pnp transistor
- T2 - BC547 npn transistor
- LED1 - 5mm LED
- D1 - 1N4148 signal diode

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 150-ohm
- R2 - 1-kilo-ohm
- R3 - 100-kilo-ohm
- R4 - 470-ohm

Capacitor:

- C1 - 0.01μF ceramic disk

Miscellaneous:

- CON1 - 2-pin connector terminal
- PZ1 - Piezo buzzer
- 5V regulated power supply

door is closed, neither the buzzer sounds nor LED1 glows. But when the door is opened, the buzzer sounds and LED1 glows, too.

Construction and testing

An actual-size, single-side PCB for the door-opening alarm circuit is shown in Fig. 2 and its component layout in Fig. 3. Enclose the PCB in a suitable small box and install it at a suitable place.

Fit the magnet on a moving part of the door and sensor MH183 on the door-frame opposite to it.

Use 5V regulated power supply for the circuit. ●



Pradeep G. is B.Sc. (Physics) and a regular contributor to international magazines. He is also a small-business owner making school/college projects in south India

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